

Linear Regression Assessment, Diagnostics & Interpretation

June 2025

Model Diagnostics and Assessment

- Does the model fit data well?
- *Should* we have fit a line -- Is model appropriate for data?
- Are predictors *really* associated with response?

Regression Conditions

Linearity

Independence

Normality of residuals

Equal variance

Flowers

Iris data set in R

- Is there a relationship between the length of an iris petal, its width and its species?

```
# Load the data  
library(datasets)  
  
# Look at the data  
glimpse(iris)
```

```
## Rows: 150  
## Columns: 5  
## $ Sepal.Length <dbl> 5.1, 4.9, 4.7, 4.6, 5.0, 5.4, 4.6, 5.0, 4.4, 4.9, 5.4, 4....  
## $ Sepal.Width <dbl> 3.5, 3.0, 3.2, 3.1, 3.6, 3.9, 3.4, 3.4, 2.9, 3.1, 3.7, 3....  
## $ Petal.Length <dbl> 1.4, 1.4, 1.3, 1.5, 1.4, 1.7, 1.4, 1.5, 1.4, 1.5, 1.5, 1....  
## $ Petal.Width <dbl> 0.2, 0.2, 0.2, 0.2, 0.2, 0.4, 0.3, 0.2, 0.2, 0.1, 0.2, 0....  
## $ Species <fct> setosa, setosa, setosa, setosa, setosa, setosa, setosa, setosa, s...
```

Our Model

```
# Use informative model names!
flower <- lm(Petal.Length ~ Petal.Width + Species,
             data = iris)

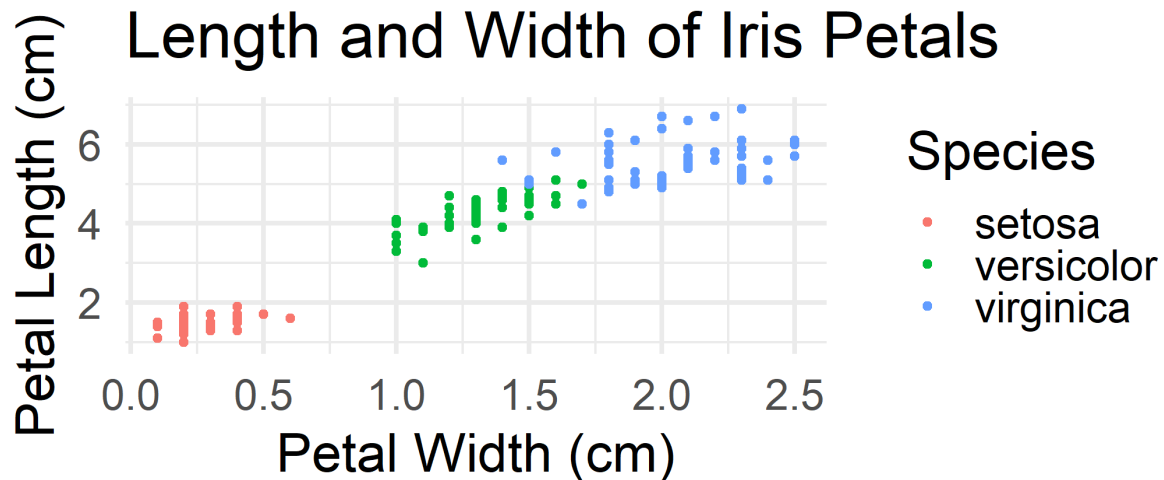
# and data should be named informatively, too!
my_iris <- iris |>
  mutate(preds = predict(flower),
         resids = resid(flower))
```

Linearity: Data Exploration

No trend, Linear trend: **OK**

```
# Visualize the data
```

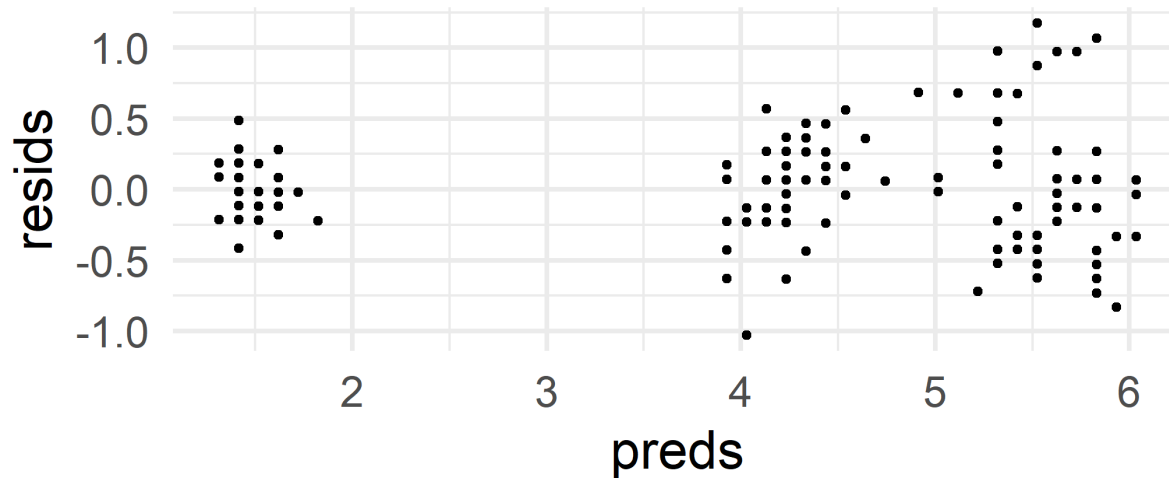
```
ggplot(iris,aes(Petal.Width,Petal.Length,color=Species)) +  
  geom_point()+  
  labs(title = "Length and Width of Iris Petals",  
        x = "Petal Width (cm)",  
        y = "Petal Length (cm)",  
        color = "Species")
```



Linearity: Residual Plots

No trends: **OK**

```
gf_point(resids ~ preds,  
         data = my_iris)
```



Your turn!

Using the multiple regression model from the ultra-processed foods data set:

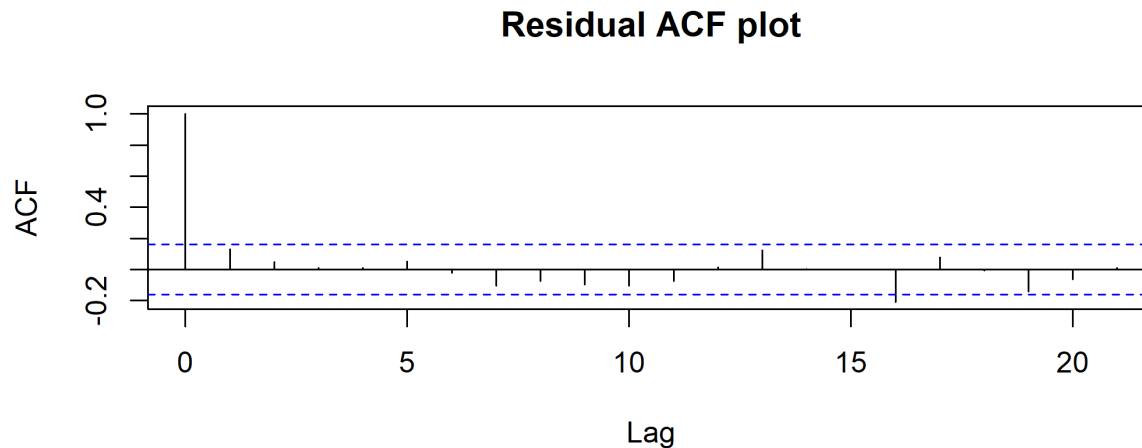
- Extract the residuals and fitted values
- Plot the response variable against one of the explanatory variables
- Plot the residuals against the fitted values

Independence of Residuals

Ponder sort order; then ACF plot

Logic and knowledge of data set

```
acf(my_iris$resids, main="Residual ACF plot")
```



Your turn!

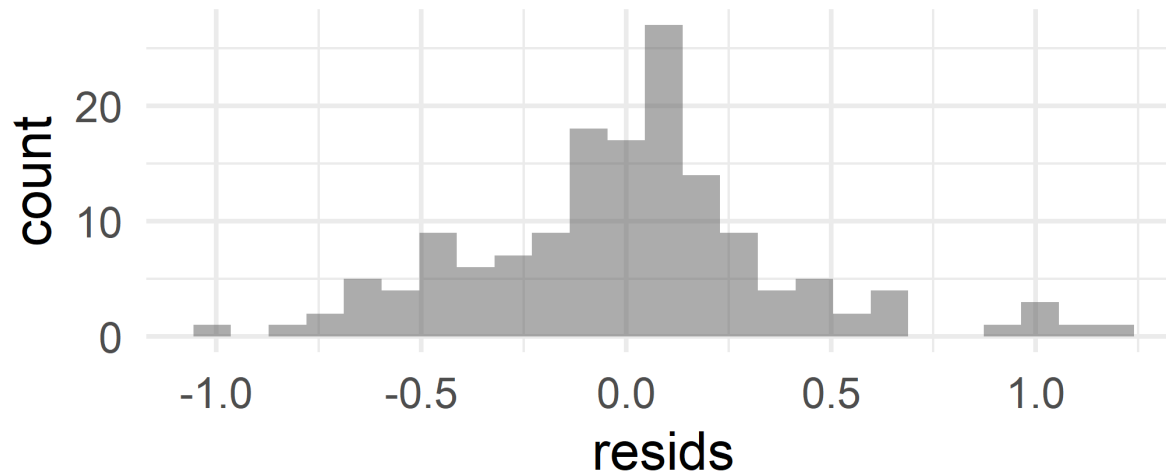
Using the same multiple regression model:

- Create an acf plot

Normality of the Residuals: Histogram

(Be quite generous)

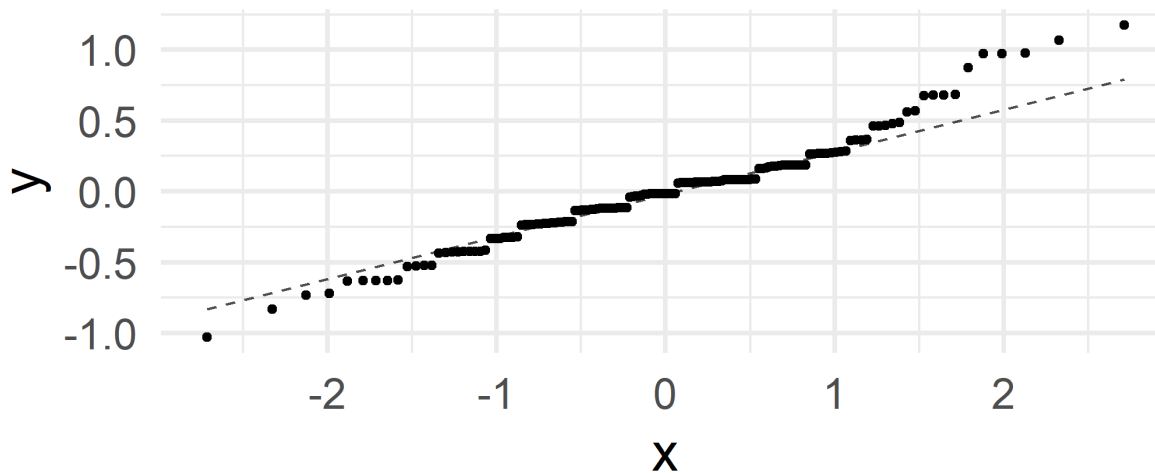
```
gf_histogram(~resids, data = my_iris)
```



Normality of the Residuals: Q-Q plot

(Be quite generous)

```
gf_qq(~resids, data = my_iris) |>  
  gf_qqline()
```



Your turn!

Using the multiple regression model from the ultra-processed foods data set:

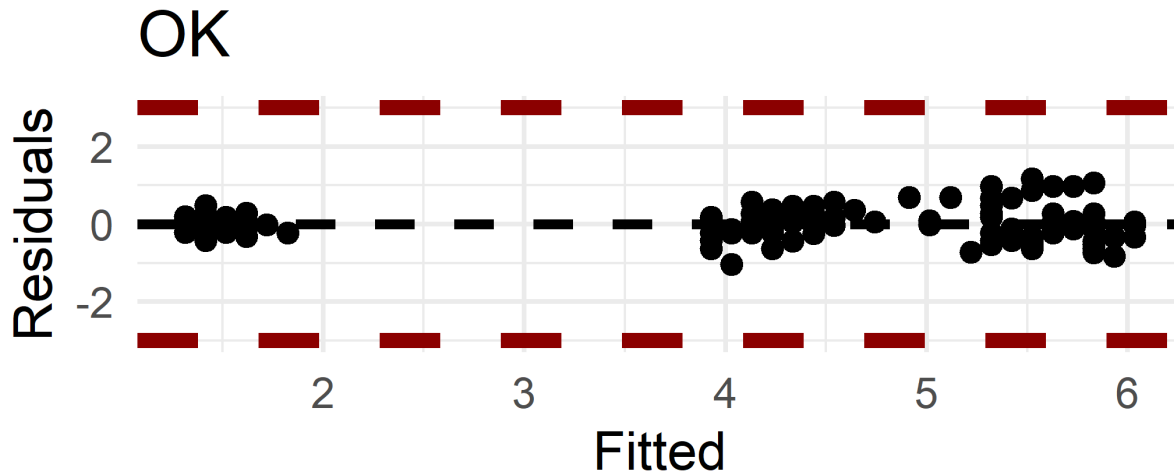
- Plot a histogram of the residuals
- Generate a QQ plot

Constant Residual Variance

Point cloud should fit well in a rectangle (no trumpet or curve)

True for residual vs. fitted *and* residual vs. any predictor

```
gf_point(resids ~ preds, data = my_iris, title = 'OK', size = 4) |>  
  gf_hline(yintercept = 3, color = 'darkred', linetype = 'dashed', size = 3) |>  
  gf_hline(yintercept = -3, color = 'darkred', linetype = 'dashed', size = 3) |>  
  gf_hline(yintercept = 0, color = 'black', linetype = 'dashed', size = 2) |>  
  gf_labs(x="Fitted", y="Residuals")
```



Your turn!

Using the multiple regression model from the ultra-processed foods data set:

- Plot the residuals against the fitted values
- Add dashed lines 3 standard deviations from the mean

Any LINE Violation -> Danger!

Conclusions can not be trusted

- slope estimates **incorrect**
- CIs and p-values **too small**
- poor prediction accuracy
- Most robust to linearity and normality
- Independence and equal variance the kickers

R^2

Simple measure of fit of model to data

NOT *goodness* of model or *appropriateness*

- Ranges from 0 - 1
 - 0: no trend
 - 1: perfect fit
- Percentage of variance explained

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Model Output

```
summary(flower)
```

```
##
## Call:
## lm(formula = Petal.Length ~ Petal.Width + Species, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.02977 -0.22241 -0.01514  0.18180  1.17449
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.21140    0.06524  18.568 < 2e-16 ***
## Petal.Width    1.01871    0.15224   6.691 4.41e-10 ***
## Speciesversicolor 1.69779    0.18095   9.383 < 2e-16 ***
## Speciesvirginica  2.27669    0.28132   8.093 2.08e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3777 on 146 degrees of freedom
## Multiple R-squared:  0.9551,    Adjusted R-squared:  0.9542
## F-statistic: 1036 on 3 and 146 DF,  p-value: < 2.2e-16
```

Interpretation

- **(Intercept)**: An *Iris setosa* with a petal width of zero has a petal length of 1.21 cm
- **Petal.Width** slope: For each cm increase in petal width, petal length also increases by a cm
- **Speciesversicolor** slope: The length of *Iris versicolor* petals are, on average 1.7 cm longer than *I. setosa*
- **Speciesvirginica** slope: The length of *Iris virginica* petals are, on average 2.3 cm longer than *I. setosa*
- The length of an iris flower petal is positively associated with its width ($\beta_1 = 1.02$, SE = 0.15), and is different between species.

$$Length_i = 1.21 + 1.02Petal.Width_i + 1.7versicolor_i + 2.3virginica_i + \epsilon_i$$

Your turn!

Using the multiple regression model from the ultra-processed foods data set:

- Interpret your model results
 - Is there an associated between the explanatory variables and the response?
 - What is the nature of the association?
 - How would you communicate your results to another scientist? To the general public?